A bounded Newton force, reversible preparations and a conditional action floor

Result, 2026-10-27; written derivation, unrefereed. A continuous bounded Newton force can send two different exact position–momentum states to the same state in finite time. Consequently it admits no reversible evolution on the original point-state space. If an admissible class of probability preparations is invariant under affine canonical maps, closed under weak limits, and has a reversible evolution agreeing with these incoming Newton trajectories, then its finite-covariance preparations have a strictly positive covariance action floor. No numerical floor is assumed in this implication.

The closure and state-covariance premises are substantive physical assumptions. An explicit classical density-ceiling model satisfies them and has a sharp floor. Thus the result selects a positive preparation scale conditionally; it does not derive quantum mechanics, a universal Planck constant, or a floor on every classical action difference.

Targeted check (GPT-6.1 Terra, 2026-10-27). Terra checked sections 1–4, including singular covariances and the classical witness, without finding a mathematical error. Its distinction between the C^1 potential and the merely continuous force is retained. The check does not validate the physical preparation premises or a quantum identification.

Retrieval: the ODE lesson constructs the Lipschitz causal limit; the self-sufficiency contrast separates Kepler incompleteness from exclusion of classical completions; the fifth-postulate note, Theorem B′ (proof read), classifies covariant Gaussian restrictions but leaves their zero threshold available. The necessity note, Theorem I (proof read), prevents treating an unread density ceiling as an automatic restriction on recorded posteriors. The present argument adds a different conditional selector: weak closure together with an actual point-state merger. No literature priority is claimed.

1. Existence without uniqueness, with no unbounded force

Fix $m, \gamma, R > 0$, $0 < \alpha < 1$, and a smooth even cutoff χ with $0 \le \chi \le 1$, $\chi = 1$ on $[-1, 1]$ and $\chi = 0$ outside $[-2, 2]$. Set

$$V(q) = -\frac{\gamma}{1+\alpha}|q|^{1+\alpha}\chi(q/R), \qquad \dot q = p/m, \quad \dot p = -V'(q). \tag{1}$$

The potential is bounded and C^1, and the force is continuous and bounded. Specifically,

$$\|V\|_\infty \le \frac{\gamma(2R)^{1+\alpha}}{1+\alpha}, \qquad \|V'\|_\infty \le \gamma(2R)^\alpha\left(1 + \frac{2\|\chi'\|_\infty}{1+\alpha}\right) =: F_*. \tag{2}$$

Inside $|q| < R$, the acceleration is $\ddot q = (\gamma/m)\,\mathrm{sgn}(q)|q|^\alpha$. Its derivative diverges at zero, so the Lipschitz hypothesis in the lesson fails. Nevertheless solutions exist globally. Here is the relevant existence argument without assuming uniqueness. Euler polygons on a fixed finite time interval have uniformly bounded momentum and position, and are equi-Lipschitz. On their common compact range the continuous vector field is uniformly continuous, so their integral-equation residual tends uniformly to zero with the mesh. A uniformly convergent subsequence therefore solves (1). The bounds

$$|p(t)| \le |p(0)| + F_*|t|, \qquad |q(t)| \le |q(0)| + \frac{|p(0)|}{m}|t| + \frac{F_* t^2}{2m} \tag{3}$$

prevent finite-time escape and allow successive continuation in either time direction. This is an existence construction by compactness, rather than the Lipschitz lesson's Cauchy construction of a unique limit.

Put

$$\beta = \frac{2}{1-\alpha} > 2, \qquad B = \left[\frac{\gamma(1-\alpha)^2}{2m(1+\alpha)}\right]^{1/(1-\alpha)}. \tag{4}$$

The curves $q(t) = \pm B(t-t_0)^\beta$, preceded by rest at zero, solve (1) while their positions remain inside $(-R, R)$. They join with two continuous derivatives at t_0. Rest and an arbitrary delay before either departure are therefore all permitted by the same initial datum $(q, p) = (0, 0)$. Direct substitution uses $\beta - 2 = \alpha\beta$ and $B^{1-\alpha} = \gamma/[m\beta(\beta-1)]$.

More decisively, choose $s > 0$ with $Bs^\beta < R$. The two different incoming states

$$z_\pm = (\pm Bs^\beta,\ \mp m\beta B s^{\beta-1}) \tag{5}$$

have the unique trajectories $q_\pm(t) = \pm B(s-t)^\beta$, $p_\pm = m\dot{q}_\pm$ for $0 \le t < s$: the vector field is locally Lipschitz wherever $q \neq 0$. Both reach $(0, 0)$ at time s. These are mergers of the full canonical state, rather than caustics of its position projection.

Point-state obstruction. No injective evolution at time s on the original (q, p) states can follow both trajectories through their arrival. A later choice of departure cannot undo the earlier merger. An extra arrival or branch label can retain the information, but is an enlargement of this state space.

2. A precise preparation theorem

Let $\mathcal{S}$ be a nonempty subset of probability measures on the canonical plane, containing at least one measure with finite covariance. Assume the following three properties.

1. **Canonical covariance of preparations.** Every affine symplectic pushforward of a member of $\mathcal{S}$ belongs to $\mathcal{S}$. In one canonical pair the linear matrices have determinant one. This is a premise about admissible states, not a claim that the particular potential (1) is invariant under all these maps.
2. **Weak closure.** A probability-valued weak limit of a sequence in $\mathcal{S}$ belongs to $\mathcal{S}$.
3. **Reversible Newton compatibility.** The force (1) admits bijective preparation evolutions $\mathcal{E}_t$ on $\mathcal{S}$, with weakly continuous state curves. Whenever a point preparation is admitted, its evolution follows its unique Newton trajectory before arrival at the cusp. No affineness assumption on $\mathcal{E}_t$ is needed for the conclusion below.

For a finite-covariance preparation write

$$\kappa(\mu) = \sqrt{\det \mathrm{Cov}_\mu(q, p)}. \tag{6}$$

This has action units. The theorem concerns covariance, including its cross term, rather than the coordinate product alone.

Theorem. Under properties 1–3,

$$0 < \kappa_* := \inf_{\substack{\mu \in \mathcal{S} \\ \mu \text{ has finite covariance}}} \kappa(\mu) < \infty. \tag{7}$$

First step: point preparations are excluded. If one Dirac measure belongs to $\mathcal{S}$, affine translations supply every Dirac measure, in particular δ_{z_+} and δ_{z_-}. Property 3 and weak time continuity then give

$$\mathcal{E}_s \delta_{z_+} = \delta_{(0,0)} = \mathcal{E}_s \delta_{z_-}, \tag{8}$$

contradicting injectivity. Thus $\mathcal{S}$ contains no Dirac measure.

Second step: arbitrarily small covariance action would supply a point. For clarity fix arbitrary calibration units $L_0, P_0 > 0$, with $S_0 = L_0 P_0$, and use dimensionless coordinates $(Q, P) =$

$(q/L_0, p/P_0)$. If (7) failed, choose preparations with dimensionless covariance matrices $C_n \geq 0$ and $\sqrt{\det C_n} = \kappa(\mu_n)/S_0 \to 0$. Subtract each mean. Choose $\epsilon_n > 0$ so that

$$\sqrt{\det D_n} \longrightarrow 0, \qquad D_n = C_n + \epsilon_n I > 0, \qquad A_n = (\det D_n)^{1/4} D_n^{-1/2}. \tag{9}$$

Such a choice follows from continuity of determinant at each C_n; it also handles rank-deficient covariances. Then $\det A_n = 1$ and

$$A_n C_n A_n^{\mathsf{T}} \leq \sqrt{\det D_n}\, I. \tag{10}$$

These dimensionless determinant-one maps represent physical canonical maps in the calibrated coordinates. By property 1 the centered, transformed preparations $\widetilde{\mu}_n$ remain admissible. For every fixed $r > 0$, Chebyshev's inequality gives

$$\widetilde{\mu}_n\{Q^2 + P^2 > r^2\} \leq \frac{2\sqrt{\det D_n}}{r^2} \longrightarrow 0. \tag{11}$$

Hence $\widetilde{\mu}_n$ converges weakly to the point measure at zero. Property 2 admits that point, contrary to the first step. The assumed finite-covariance member also makes the infimum finite. □

No calibrated numerical value is hidden in L_0, P_0: their positive values are arbitrary and (7) is unchanged by changing them. The state class supplies the value of κ_*. The theorem does not establish that the infimum is attained, that Gaussian states are allowed, or that all states above the floor are allowed. It needs neither a Gaussian ansatz nor closure under adding Gaussian noise.

One may define $\hbar_{\mathrm{prep}} = 2\kappa_* > 0$ as a candidate reduced action scale. Identifying it with the phase constant $\hbar$ needs a separate physical argument. In quantum mechanics the familiar Robertson–Schrödinger form is $\det \mathrm{Cov}(q, p) \geq \hbar^2/4$; the equality of forms alone proves no identification. The full Planck constant would be $h = 2\pi\hbar$, not $\hbar$.

3. Which assumptions do the work?

Without weak closure, excluding exact points leaves preparations with arbitrarily small covariance. For example, the class of all absolutely continuous phase-space probabilities is canonically invariant, has no Dirac members, and contains arbitrarily narrow Gaussians. Its weak closure includes points. Thus the conclusion is not obtained merely by replacing points with distributions.

Without canonical squeezing, a class supported on a fixed line with a fixed nonzero extent may be weakly closed and contain no point while having zero covariance determinant. Canonical covariance rules out that degenerate evasion. Without the cusp-compatibility demand, all probabilities are weakly closed and canonically invariant and contain points; free and harmonic dynamics transport them reversibly.

Weak closure is not a consequence of finite experimental resolution. It is an additional completion rule admitting concentration limits whenever they are weak probability limits. Requiring it while retaining all arbitrarily concentrated classical preparations is exactly what would reintroduce the forbidden point. Whether nature supplies this closure, the full canonical preparation class, and the exact cusp force is open. Restricting all forces to smooth Lipschitz laws removes this particular selector.

4. A classical model satisfies all three premises

Fix a finite density ceiling $M > 0$, of reciprocal-action units, and let

$$\mathcal{S}_M = \{\mu : \; d\mu = \rho\, dq\, dp, \quad 0 \leq \rho \leq M, \quad \int \rho\, dq\, dp = 1\}. \tag{12}$$

This class is weakly closed: if its probabilities converge weakly, their limiting measure satisfies $\int \phi \, d\mu \le M \int \phi \, dq \, dp$ for every nonnegative compactly supported continuous ϕ, which gives the same density bound. It is invariant under affine symplectic maps because they preserve canonical area.

For (1), the energy-zero set is the union of the two graphs $p = \pm\sqrt{-2mV(q)}$ and has canonical area zero. Outside it the Newton trajectory is unique and complete. Negative-energy paths cannot reach zero; positive-energy paths cross it with nonzero velocity and are uniquely continued by $\dot{q} = \pm\sqrt{2(E - V(q))/m}$ near the crossing. Away from zero the field is smooth. This full-area flow preserves canonical area: where $p \neq 0$, energy–time coordinates give $dq \, dp = dE \, dt$ up to orientation, and the flow translates t; turning points away from zero have the usual smooth Hamiltonian charts. The resulting almost-everywhere group pushes $\mathcal{S}_M$ bijectively to itself. Its state curves are weakly continuous by bounded convergence along the continuous trajectories. No member charges the exceptional zero-energy set, and no Dirac preparation is admitted. Property 3 is therefore satisfied, including its stated point-compatibility condition.

The floor can be computed sharply:

$$\inf_{\mu \in \mathcal{S}_M \text{ with finite covariance}} \sqrt{\det \mathrm{Cov}_\mu(q,p)} = \frac{1}{4\pi M}. \tag{13}$$

To check it, center and canonically isotropize a positive covariance as in section 2, now without regularization. In the calibrated plane the density cap is MS_0. Among probabilities with this cap the minimum of $\mathbb{E}(Q^2+P^2)$ is attained by filling the disk about zero at maximum density: moving mass from a larger radius into any unfilled smaller radius reduces the expectation. The disk has radius $a_0^2 = 1/(\pi M S_0)$ and mean square radius $a_0^2/2$. Isotropy gives $2\kappa/S_0 \ge 1/(2\pi M S_0)$, proving the bound. A uniform canonical ellipse, the image of this disk, attains equality. A singular covariance would concentrate a density on a line and is impossible in (12).

Thus the conditional theorem is consistent and genuinely weaker than quantum kinematics. Its classical witness already carries a supplied scale $1/M$; the theorem says that some positive scale must appear under its hypotheses, not how nature chooses it. It is an unread-state model. Unrestricted classical conditioning can violate its ceiling, as the necessity note's two-pointer construction proves. Claiming an operational completion would also require a physical law of instruments and records preserving its allowed states.

5. A supplied positive $\hbar$ gives a complete quantum comparison

For every $\hbar > 0$, the same bounded real potential has

$$H_\hbar = -\frac{\hbar^2}{2m}\frac{d^2}{dq^2} + V(q), \qquad D(H_\hbar) = H^2(\mathbb{R}). \tag{14}$$

The free operator is self-adjoint on H^2. Multiplication by V is bounded and symmetric, with relative bound zero. The Kato–Rellich theorem therefore makes (14) self-adjoint on this same domain, with $H_\hbar \ge -\|V\|_\infty$. Its spectral calculus supplies the unique strongly continuous unitary group $U_\hbar(t) = \exp(-itH_\hbar/\hbar)$ for every real time. These are [@Teschl2009, Theorem 6.4, printed p.135, and Theorem 5.1, pp.123–124; passages read in the held first edition], with the free domain in Theorem 7.8, pp.167–168. See the source companion.

This comparison needs neither an unbounded collision force nor a choice of singular boundary condition. It still supplies $\hbar > 0$; the classical model of section 4 prevents inferring that this is the only completion. The quantum state space also is not a class of positive joint canonical probabilities with unrestricted classical conditioning. No such identification was used in section 2. No relativistic premise enters, so no c is needed.

6. Consequence for STATE

The link from Lipschitz control to Newton's polygons now has a complement: continuous bounded forces suffice for polygon existence but can merge full point states, obstructing reversibility there. Weak preparation closure and canonical covariance convert that obstruction into a conditional positive covariance action scale. The density-ceiling control leaves quantum identification and physical record closure open. Next test whether a record/composition principle can independently justify or replace these closure premises, rather than assuming a numerical floor or discarding classical enlarged-state completions.